**Habit:** Mostly perennial herbs persisting by means of a sympodial rhizome (*Polygonatum)*, by a bulb (*Lilium*) corm (*Colchicum*), shrubby or tree like (*Yucca* and *Dracaena*). Woody climbers, climbing with the help of stipular tendrils in *Smilax*. Trees in (*Xanthorrhoea),* succulents (*Aloe)*.

**Root:** Adventitious and fibrous, and typically contractile.

**Stem:** Stems usually bulbous, rhizomatous in some, aerial, erect (*Dracaena*) or climbing (*Smilax)* in *Ruscus* the ultimate branches are modified into phylloclades, In *Asparagus* stem is modified into cladodes and the leaves are reduced to scales.

**Leaf:** Leaves are radical (*Lilium*) or **cauline** (*Dracaena*), usually alternate, opposite (*Gloriosa*), sometimes fleshy and hollow, reduced to scales (*Ruscus* and *Asparagus*). The venation is parallel but in species of *Smilax* it is reticulate. Leaves are usually exstipulate, but in *Smilax*, two tendrils arise from the base of the leaf, which are considered modified stipules.

**Inflorescence:** Flowers are usually borne in simple or branched racemes (*Asphodelus*) spikes in *Aloe*, huge terminal panicle in *Yucca*, solitary and axillary in *Gloriosa*, solitary and terminal in *Tulipa.*

**Flowers:** Flowers are often showy, pedicellate, bracteate, ebracteolate, except *Dianella* and *Lilium*, bisexual, actinomorphic, trimerous, hypogynous, rarely unisexual (*Smilax*) and are dioecious, rarely tetramerous (*Maianthemum*), slightly zygomorphic (*Lilium*) and hypogynous.

**Perianth:** Tepals 6 biseriate arranged in two whorls of 3 each, apotepalous or rarely syntepalous as in *Aloe.* Usually petaloid or sometimes sepaloid, odd tepal of the outer whorl is anterior in position, valvate or imbricate, tepals more than six in *Paris quadrifolia.*

**Androecium:** Stamens 6, arranged in 2 whorls of 3 each, rarely stamens are 3 (*Ruscus*), 4 in *Maianthemum,* or up to 12, apostamenous, opposite to the tepals, sometimes epitepalous; filaments distinct or connate, anthers dithecous, basifixed or versatile, extrose, or introse, dehiscing usually by vertical slit and sometimes by terminal pores; rarely **synstamenous** (*Ruscus*).

**Gynoecium:** Tricarpallary, syncarpous, the odd carpel usually anterior, ovary superior, trilocular, with 2 rows of numerous ovules on axile placextation; rarely unilocular with parietal placentation, style usually one; stigmas 1 or 3; rarely the ovary is inferior (*Haemodorum*), nectar – secreting **septal glands** are present in the ovary.

**Fruit:** Fruit usually a septicidal or loculicidal capsule or a berry as in *Asparagus* and *Smilax.*

**Botanical description of *Allium cepa***

(In APG classification, *Allium cepa* is placed under the family Amaryllidaceae)

**Habit:** Perennial herb with bulb.

**Root:** Fibrous adventitious root system

**Stem:** Underground bulb

**Leaf:** A cluster of radical leaves emerges from the underground bulb, cylindrical and fleshy having sheathy leaf bases with parallel venation.

**Inflorescence: Scapigerous** i.e. the inflorescence axis (peduncle) arising from the ground bearing a cluster of flowers at its apex. Pedicels are of equal length, arising from the apex of the peduncle which brings all flowers at the same level.

**Flower:** Small, white, bracteate, ebracteolate, pedicellate, complete, trimerous, actinomorphic and hypogynous. Flowers are protandrous.

**Perianth:** Tepals 6, white, arranged in two whorls of three each, syntepalous showing valvate aestivation.

**Androecium:** Stamens 6, arranged in two whorls of three each, epitepalous, apostamenous /free and opposite to tepals.



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